- We can store point C by using a single real value (the distance from the line passing through the endpoints);
- The tangent to the curve that's parallel to segment SE is exactly halfway between the same segment and point C.

We'll later see why the third point is especially important for us.

Before that, we need to give a brief overview of the methods that can be used to check intersections between Bézier curves.

- **Bézier subdivision** is a method based on *convex hull*[27]: for a Bézier curve with n-2 control points, its convex hull is the polygon with n sides whose vertices are the endpoints and control points. Figure 15.24 illustrates how it's possible to compute the intersection of two curves by comparing their convex hulls. If they do not overlap, the curves do not intersect; otherwise, the curves are each split in half and the two halves of one curve are checked for overlap against the two halves of the other curve.

  This step can be iterated, until either there is no overlap between any pair of sections, or the curves are split into sections so small that can be approximated with segments (within a certain acceptable error): then, we can just use algorithm in section 15.4.1 to check that no pairs of segments intersect.